An Explainable Hybrid of Lexicon Rules and Multilingual Transformers for Vietnamese Hate Speech Detection

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Abstract. Fine-tuned multilingual transformers dominate Vietnamese hate speech detection, yet they are opaque and brittle on the surface forms moderation sees: teencode, dropped diacritics, inserted separators and abbreviated profanity. We ask whether a lexicon rule engine can be attached to such a transformer as an *evidence channel* rather than a competing classifier. On the ViHSD test set (6,680 comments, 83% CLEAN) we fuse a 668-term Vietnamese hate-speech lexicon, matched through accent-less, leet, repeat-collapsed and separator-stripped views, with four backbones that differ only in their optimisation recipe. The headline effect is small and not distinguishable from zero: the focused backbone improves from 0.6575 to 0.6602 macro-F1, a net of 8 comments (23 fixed, 15 broken), with a paired-bootstrap $p = 0.29$. The value we report is not the delta but its *provenance*. The rule channel fires on 15.75% of test comments with 58.75% precision, and all 23 corrections are rule-consistent: 16 HATE recoveries where the rule fires in the corrected direction, and 7 CLEAN de-flagging cases where it is correctly silent. All 10 rule-driven regressions push towards HATE, localising the failure mode to an aggressive channel rather than to lexical coverage. We further identify 132 missed corrections where the rule already carries directional evidence fusion does not act on. These diagnostics, not the fourth decimal of macro-F1, are the contribution.

Keywords: Vietnamese hate speech · ViHSD · Explainable AI · Hybrid fusion · Lexicon rules · Error analysis.

1 Introduction

Vietnam has one of the fastest-growing social-media user bases in Southeast Asia and the volume of Vietnamese text platforms must moderate keeps rising. The language remains low-resource, and its social-media register is hard: writers abbreviate profanity into *teencode* (`dm`, `cc`, `vl`), drop diacritics, stretch letters, insert separators inside words, and rely on irony or quotation to be hostile without writing a dictionary slur.

The release of UIT-ViHSD [15], roughly 33k comments labelled CLEAN, OFFENSIVE or HATE, made supervised transformer fine-tuning the default approach. Monolingual encoders such as PhoBERT [19] and ViSoBERT [21], multilingual DeBERTa-v3 variants

[10,9,18] and task-specific systems [23] all perform well, yet on ViHSD strong encoders plateau: about 83% of test comments are CLEAN, HATE and OFFENSIVE overlap in annotation, and their boundary is often a matter of intensity rather than lexical content.

Two problems follow. The first is *opacity*: a decision that removes a comment needs a reason a moderator or appeals reviewer can inspect, yet post-hoc attribution such as LIME [25] or SHAP [14] approximates the model rather than describing the decision procedure. The second is *brittleness on surface form*: a transformer may never see `đm` written as `d.m`, `đ*m` or `địt mẹ`, while a lexicon can be written to match all of them explicitly.

We ask a narrow question: *can a lexicon-based rule engine be attached to a transformer as an evidence channel, so that every decision carries inspectable lexical support, without degrading accuracy?* We do not claim that the hybrid is more accurate: on the ViHSD test split the focused backbone improves by 0.0027 macro-F1, a paired-bootstrap p of 0.29, a net of eight comments out of 6,680. We claim something different and, for deployment, more useful: the small change is *attributable*. Every one of the 23 corrections and 15 regressions traces to a specific mechanism, and the trace exposes a systematic failure that aggregate metrics hide.

We fuse a 668-term Vietnamese hate-speech lexicon with four mDeBERTa-v3 backbones that differ only in their optimisation recipe. The rule engine normalises the comment, builds accent-less, leet, repeat-collapsed and separator-stripped views, scores each matched span by severity, entry confidence, match type and context, and aggregates matches per label; fusion mixes each channel and applies frozen per-channel thresholds.

The rule channel fires on 15.75% of test comments at 58.75% precision, supports all 23 corrections and drives all 10 rule-driven regressions; a further 132 *missed corrections* show the rule pointing right while the CLEAN channel blocks the fix.

The contributions are: (i) a frozen, reproducible hybrid protocol in which the rule engine is an independent, inspectable channel rather than a competing classifier; (ii) a per-decision provenance analysis assigning every changed decision to a rule-driven or threshold-driven mechanism; (iii) a behavioural audit of the rule channel with a multi-label cue taxonomy of the recovered HATE cases; (iv) the identification of 132 missed corrections, which motivates evidence-gated rather than score-averaged fusion; and (v) five case studies with matched spans and per-channel scores.

2 Related Work

Lexicon-based detection. Pre-transformer detectors used lexicons and hand-written surface features [28]: Gitari et al. [8] score a comment from a curated profanity lexicon and apply rules over the matched phrases—inspectable, but hand-curated, with recall bounded by the list. Wiegand et al. [29] automate that induction step, the closest antecedent to how our entries acquire severity and confidence. Schmidt and Wiegand's survey [27] explains the move to learned models: the HATE/OFFENSIVE boundary is combinatorial, not lexical. Davidson et al. [4] found the two classes share most of their surface vocabulary—the tension that makes our channel over-fire on OFFENSIVE—and surveys [7] list the recurring difficulties: imbalance, subjectivity, overlap, obfuscation, domain shift. We take these as the object of analysis rather than as obstacles to a classifier.

Vietnamese resources and models. ViHSD [15] is the reference three-way benchmark, ViCTSD [20] provides related data, and ViCM [24] adds code-mixed comments. PhoBERT [19] adapts RoBERTa to Vietnamese, ViSoBERT [21] is pre-trained on social media, and ViHateT5 [23] casts the task as generation. ViHOS [11] annotates the *spans* that make a comment hateful, the closest Vietnamese work to our notion of evidence; our spans come from a lexicon matcher rather than human annotation, but the motivation is the same. SMTCE [22] standardises Vietnamese social-media evaluation, which is why we report macro-F1 rather than accuracy alone.

Multilingual transformers. mBERT [5] and XLM-R [3] established multilingual pre-training; DeBERTa [10] introduced disentangled attention and DeBERTa-v3 [9] adds ELECTRA-style pre-training, whose `microsoft/mdeberta-v3-base` checkpoint [18] we fine-tune.

Explainability and hybrid systems. HateXplain [16] made human rationales and explanation quality benchmark quantities. LIME [25] and SHAP [14] are model-agnostic but do not guarantee that the highlighted tokens are the ones the decision used. Our rule channel is part of the decision function, so its matched spans are *faithful by construction* for the lexical component, while the neural component remains opaque. We differ in two ways: we fix the accuracy question with a paired test and analyse provenance instead, and we report the rule channel's own error profile and the direction of every rule-induced change. To our knowledge, this per-decision provenance audit has not been reported for Vietnamese hate speech.

3 Data and Lexicon

3.1 The ViHSD corpus

We use UIT-ViHSD [15], a corpus of Vietnamese social-media comments annotated with three mutually exclusive labels: CLEAN (no offensive content), OFFENSIVE (offensive language that does not target a protected group as hate speech) and HATE (hateful or seriously derogatory content). The OFFENSIVE/HATE distinction depends on target and intensity rather than on a particular word, so a purely lexical rule engine is insufficient on its own.

After cleaning, the corpus contains 33,398 comments. We use the fixed split shipped with the project, and every system—rule engine, backbone, fusion variant and analysis script—reads the same files. Each run directory stores the SHA-256 digest of the split it used and re-checks it before training, ruling out a silently re-shuffled split. Training contains 24,046 comments (2,556 HATE, 1,605 OFFENSIVE, 19,885 CLEAN), validation 2,672 (270/212/2,190) and test 6,680 (688/444/5,548).

The test split is imbalanced: 83.05% CLEAN, 10.30% HATE and 6.65% OFFENSIVE. Accuracy is therefore a poor headline metric—a constant CLEAN predictor already reaches 0.8305—so macro-F1 is primary, always reported with per-class F1. The 6,680 test rows contain 6,571 distinct texts, i.e. 109 duplicated rows. They are kept because removing them would change the frozen test set; one appears twice among the 15 regressions, which therefore cover 14 distinct texts.

3.2 The hate-speech lexicon

The rule engine is driven by a Vietnamese hate-speech lexicon of **668 entries**, mined and labelled from the *training* portions of three corpora: ViHSD; R-ViHSD, a cleaned release that adds TDTU YouTube and voz-forum comments to ViHSD [2]; and the code-mixed ViCM corpus [24]. The three training splits yield 1,062 candidate terms, of which 257, 665 and 140 are first attested in ViHSD, ViCM and R-ViHSD; labelling reduces them to the entries kept below. No comment from any evaluation split contributes to the dictionary, and each entry records which corpora supported it (414 of the 668 occur in all three). Each entry stores the surface term and a canonical form, a lexical label (HATE or OFFENSIVE) with a per-label severity, an entry confidence estimated from the training data, a context-required flag, and a review status (*verified* or *auto*); both statuses may match, but the flag is retained in the evidence record.

The lexicon is deliberately not a list of slurs. Most entries—551 of 668 (82.5%)—are multi-word constructions such as *mả cha*, *con mẹ* or *ngu hơn*, and 524 (78.4%) are context-required, because Vietnamese insults frequently work by comparison and implication rather than by a single taboo word. Only 74 entries carry an OFFENSIVE lexical label against 594 for HATE, which is why the rule engine is weak on OFFENSIVE and fusion keeps that channel mostly neural. Three properties make the lexicon worth keeping: it is *editable* (a new variant needs no retraining), *deterministic* (the same comment yields the same spans) and *inspectable* (each score decomposes into a term, a match type and a context factor).

Ethics of examples. All quoted comments are public ViHSD test comments. Because they contain profanity and slurs, we mask the offending characters with ∗ in every table; evaluation always runs on the unmasked text. No user identifiers are reproduced, and names that occasionally appear inside comments are part of the public corpus text.

4 Method

The hybrid has three parts: a rule engine that produces lexical scores and spans, a transformer that produces class probabilities, and a fusion layer that combines them channel by channel (Fig. 1). The branches share no parameters, which allows the rule branch to be audited on its own.

4.1 The lexical rule engine

Normalisation and views. The comment is Unicode-normalised to NFC, invisible characters are removed, whitespace is collapsed, character runs are shortened to at most three, and the text is lower-cased while keeping Vietnamese diacritics. Normalisation returns an *offset map* to the raw comment, so a match on a cleaned view is always reported as a span of the original text. Auxiliary views handle the writing tricks that matter: accent-less, leet, repeat-collapsed, and a compact view with separators removed (`n g u`). The matcher records, for every hit, the dictionary term, label, view, raw span and a *match type* (*exact*, *accentless_alias*, *leet*, *repeat_collapsed*, *separator_inserted*, *obfuscated_combined* or *fuzzy*).

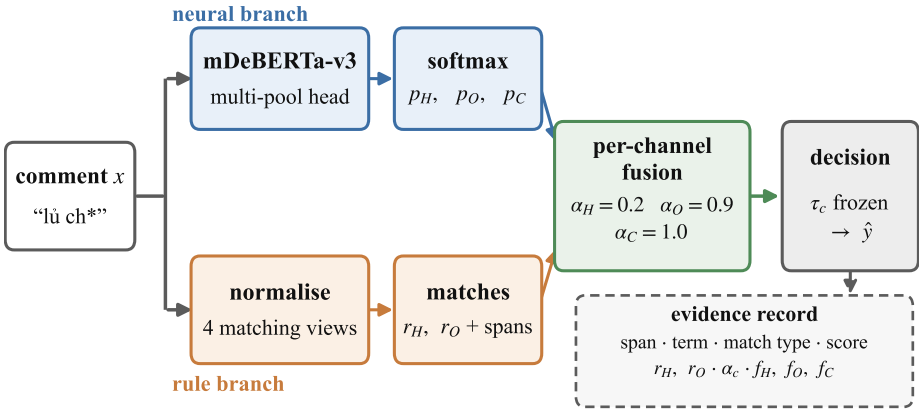


Fig. 1. The hybrid. A comment enters two independent branches. The neural branch produces class probabilities; the rule branch produces per-label lexical scores and the spans that produced them. Fusion mixes each channel with its own weight α_c, and frozen per-channel thresholds turn the three scores into a label. The evidence record is emitted with the decision and makes the lexical part of the decision inspectable. Weights and thresholds shown are the values selected on validation for the focused backbone.

Scoring. Every match i receives

$$s_i \;=\; \sigma_{\ell(i)} \cdot c_i \cdot m_i \cdot \kappa_i, \tag{1}$$

where $\sigma_{\ell(i)}$ is the severity of the matched label (0.95 for HATE, 0.60 for OFFENSIVE after tuning), c_i the entry confidence from the training split, m_i the match-type factor and κ_i a context factor for quotation, negation and context-required terms. Matches of one label are de-duplicated and aggregated with a saturating sum over at most $K = 10$ matches:

$$r_\ell \;=\; 1 - \prod_{i \in T_\ell} (1 - s_i)\,. \tag{2}$$

For fusion they become a three-channel pseudo-probability

$$\tilde{\mathbf{r}} \;=\; \frac{1}{1 + \min(r_H, r_O)} \begin{bmatrix} r_H \\ r_O \\ 1 - \max(r_H, r_O) \end{bmatrix}, \tag{3}$$

which preserves the ordering of the two toxic channels while placing CLEAN on the same scale. When the rule is silent, $\tilde{\mathbf{r}} = (0, 0, 1)$: no evidence is treated as maximal cleanliness evidence, the assumption that Sect. 7 tests.

Tuning. Scoring weights are tuned on validation by greedy coordinate descent, with thresholds re-optimised for every candidate; test is never loaded. The search made the engine more aggressive: severity HATE rose to 0.95, the accentless-alias factor was driven to 0.0, the repeat-collapsed factor to 0.4, the combined-obfuscation factor to 0.3, and the

negation and context-required factors to 1.0. Validation preferred *more* matches over *cautious* ones: the engine is a narrow, high-threshold detector, not a comprehensive obfuscation detector.

4.2 The neural branch

We fine-tune `microsoft/mdeberta-v3-base` [9,18]. The focused backbone (SENSITIVEAI-VI-CUSTOM) uses a multi-pool head: the final hidden states are pooled with the `[CLS]` token, mean pooling and max pooling, the three views are concatenated, projected to a 512-dimensional bottleneck with dropout 0.1, and mapped to three logits. Training uses cross-entropy, AdamW [13], a linear schedule with 10% warm-up, learning rate 3×10^{-5}, effective batch size 32, maximum length 256 and at most 30 epochs, with early stopping on validation macro-F1; the best checkpoint is frozen before test. Three further backbones use the same encoder with different recipes: a standard-head baseline, a two-stage model with discriminative learning rates and focal loss [12], and a three-stage curriculum [1] with manual class weights.

4.3 Fusion and decision

Let p_c be the transformer probability for class c and $\tilde{r}_c$ the renormalised rule score. Per-channel fusion is

$$f_c \;=\; \alpha_c\, p_c \;+\; (1-\alpha_c)\,\tilde{r}_c, \qquad c \in \{\text{HATE}, \text{OFFENSIVE}, \text{CLEAN}\}. \tag{4}$$

A single global α is the special case $\alpha_H = \alpha_O = \alpha_C$. A class is *accepted* when $f_c \geq \tau_c$; the accepted class with the highest f_c wins, and if none is accepted the decision falls back to $\arg\max_c f_c$.

We compare four families: global α with argmax, global α with thresholds, a multinomial logistic meta-classifier over the six AI and rule scores, and per-channel α with thresholds. All tunable quantities are selected on validation and frozen before the single test pass. The per-channel variant is primary because the classes have different rule behaviour: the rule channel is precise on HATE, weak on OFFENSIVE and uninformative on CLEAN, so one α forces a bad compromise. Validation discovers the asymmetry: $\alpha_H = 0.2$, $\alpha_O = 0.9$, $\alpha_C = 1.0$, with $\tau_H = 0.14$, $\tau_O = 0.66$, $\tau_C = 0.60$. The HATE channel is 80% lexical, OFFENSIVE 90% neural and CLEAN purely neural; the interaction between α_H and τ_H drives Sect. 7.

4.4 Evidence record and protocol

For every comment the system emits, with the label, an evidence record: matched spans in the raw text, dictionary term and label of each span, matching view and match type, per-match score, aggregated rule scores, per-channel weights and the three fused scores. Given the record and the transformer probabilities, a reviewer can recompute the decision exactly, which is what we mean by *faithful by construction* for the lexical channel; the neural contribution remains a black box, and we do not claim otherwise. Every tunable quantity is frozen before the single test pass, and the error analysis is descriptive and performed after it.

5 Experimental Setup

5.1 Backbones and tuning

Table 1 lists the four backbones. All use `microsoft/mdeberta-v3-base`, the same tokenizer, split and evaluation code, and differ only in the head and optimisation recipe. The focused model, SENSITIVEAI-VI-CUSTOM, is chosen because it gives the clearest positive net effect (23 corrections against 15 regressions), not because it has the highest standalone score: the strongest backbone, SENSITIVEAI-VI-DLR2STAGE, reaches 0.6638 macro-F1 but is almost indifferent to fusion (3 corrections, 2 regressions), which makes it a poor subject for a study of what fusion changes.

Table 1. The four backbones and the paired tests between AI-only and the hybrid. All backbones share the encoder (`microsoft/mdeberta-v3-base`), tokenizer, split and evaluation code and differ only in the optimisation recipe. $\Delta = \text{AI} - \text{hybrid}$ in macro-F1, so a negative value favours the hybrid; the 95% interval is the paired-bootstrap percentile interval over 10,000 resamples. b is the number of comments the AI gets right and the hybrid gets wrong, c the reverse; the last column is the exact McNemar p.

Backbone (recipe)	AI	Hybrid	Δ	95% CI	p_{boot}	b/c	$p_{\text{McN.}}$
SENSITIVEAI-VI (std. head)	0.6545	0.6494	$+0.0051$	$[-0.0020, 0.0120]$	0.1592	61/30	**0.0015**
SENSITIVEAI-VI-CUSTOM (multi-pool)	0.6575	**0.6602**	-0.0027	$[-0.0079, 0.0027]$	0.2942	15/23	0.2559
SENSITIVEAI-VI-DLR2STAGE (2-stage)	**0.6638**	0.6637	$+0.0000$	$[-0.0020, 0.0024]$	0.9782	2/3	1.0000
SENSITIVEAI-VI-CURRICULUM (3-stage)	0.6590	0.6618	-0.0028	$[-0.0058, 0.0004]$	0.0868	8/31	**0.0003**

Rule-engine weights are tuned by greedy coordinate descent over the scoring parameters, with the three decision thresholds re-optimised on validation macro-F1 over a 101-point grid at every candidate; the search never touches the lexicon. Fusion follows the same discipline: thresholds by coordinate descent over the same grid for three rounds, per-channel weights by coordinate ascent from the AI-only corner, and a multinomial logistic meta-classifier fitted on the six validation features. All selected values are frozen before the test pass.

5.2 Metrics and implementation

Macro-F1 is primary (Sect. 3.1). We report accuracy and per-class precision, recall and F1, and evaluate differences with two paired tests: a paired bootstrap over comments (10,000 resamples) for the macro-F1 difference [6], and an exact McNemar test on the correctness contingency table [17]. The bootstrap asks whether the aggregate metric moved; McNemar asks whether the pattern of individual decisions changed.

The transformer uses PyTorch and Hugging Face Transformers [30]; metrics follow the scikit-learn definitions and are computed by one shared evaluator [26]. The rule engine is pure Python, runs on CPU, and processes a comment in 0.20 ms on average (median 0.15 ms), roughly 4,900 comments per second, so it can run on every comment rather than as a sampled audit.

5.3 Reproducibility

The split is hashed and re-checked, the rule-engine evaluation directory is recorded in the fusion metrics and re-read rather than recomputed, and the analysis script asserts that

the rule scores shipped with the fusion predictions match the frozen tuned output. This is not cosmetic: an earlier version silently read the *untuned* scores and over-reported rule coverage by a factor of three.

6 Results

6.1 The hybrid does not improve accuracy

Table 2 reports every decision variant for the focused backbone and Fig. 2(a) plots the same ordering. The AI alone reaches 0.6575 macro-F1 and 0.8762 accuracy; the primary per-channel hybrid reaches 0.6602 and 0.8774, a gain of 0.0027 macro-F1 and eight additional correct comments (5,861 versus 5,853). The gain is concentrated in HATE: F1 rises from 0.5830 to 0.5906, driven by recall (0.5334 to 0.5494) at a small cost in precision (0.6427 to 0.6385). OFFENSIVE and CLEAN do not pay for it (0.4493 to 0.4494, 0.9403 to 0.9407): the change is an extension of HATE recall, not a reallocation between classes.

The other variants locate the effect: per-channel weights add 0.0010 macro-F1 over a single global weight (0.6583 to 0.6593) and thresholds a further 0.0009. The logistic meta-classifier is *worse* than the AI alone (0.6513): fitting six features on 2,672 validation comments overfits the rule scores and shifts the operating point towards HATE recall. Thresholds do make the rule engine usable on its own—0.5258 macro-F1, identical standalone and inside the pipeline, against 0.4714 with argmax, which collapses on OFFENSIVE—but it stays far below the transformer (0.8421 accuracy).

Table 2. Decision-level variants for the focused backbone (SENSITIVEAI-VI-CUSTOM) on the test split. Every variant reads the same frozen scores; only the mixing and decision rule differ. The last row is the paper's primary system. *Per-class columns are F1*; the best macro-F1 is in bold. The two rule-only rows are scored inside the fusion pipeline on the renormalised rule scores; the standalone rule engine, with its own thresholds, is reported in the text.

System	Acc.	Macro-F1	HATE	OFFENSIVE	CLEAN
AI only (argmax)	0.8762	0.6575	0.5830	0.4493	0.9403
Rule only (argmax)	0.8494	0.4714	0.4867	0.0000	0.9274
Rule only (frozen thresholds)	0.8421	0.5258	0.4102	0.2389	0.9284
Fusion, global α (argmax)	0.8769	0.6583	0.5851	0.4490	0.9408
Fusion, global α (thresholds)	0.8769	0.6583	0.5851	0.4490	0.9408
Fusion, logistic meta-classifier	0.8626	0.6513	0.5818	0.4379	0.9341
Fusion, per-channel α (argmax)	0.8774	0.6593	0.5882	0.4488	0.9408
Fusion, per-channel (thresholds)	0.8774	**0.6602**	0.5906	0.4494	0.9407

6.2 The difference is not distinguishable from zero

Table 1 makes the negative result explicit. For the focused backbone the paired-bootstrap difference is -0.0027 with a 95% interval of $[-0.0079, +0.0027]$ and $p = 0.29$; McNemar on the 38 changed decisions gives $b = 15$, $c = 23$, $p = 0.26$. Across the four backbones every bootstrap interval contains zero. Where McNemar does reach significance the sign flips: the standard-head backbone loses 61 comments to the hybrid and wins back 30 ($p = 0.0015$), while the curriculum backbone loses 8 and wins back 31

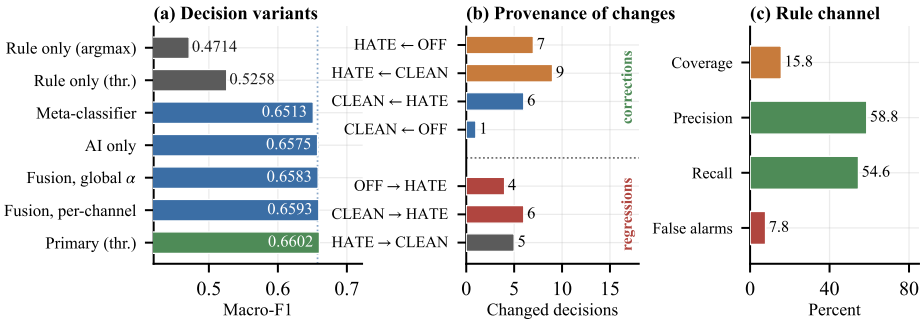


Fig. 2. Quantitative summary for the focused backbone. (a) Macro-F1 of every decision variant; the dotted line is AI-only and the green bar is the primary system. (b) Provenance of the 38 changed decisions: the four correction types and the three regression types. (c) Behaviour of the rule channel on the test split: coverage, precision when firing, recall on non-CLEAN comments, and false alarms on CLEAN. Values are percentages; raw counts are in the text and in panel (b).

($p = 0.0003$). Fusion therefore helps one recipe and hurts another by margins the aggregate macro-F1 cannot separate. We do not claim an accuracy improvement: the hybrid is *accuracy-neutral*, changing 38 decisions without moving the aggregate metric beyond noise.

6.3 The rule channel is narrow and precise-ish

The rule engine fires on 1,052 of 6,680 comments (15.75%), not on the majority of the test set. When it fires, 58.75% of the comments are truly non-CLEAN, so roughly two in five firings are false alarms; it recovers 54.59% of non-CLEAN comments and fires on 7.82% of CLEAN comments (Fig. 2c). It is useful as evidence, not as a decision: silent on 45% of toxic comments and wrong about a substantial share of those it flags. Coverage is low by construction—most entries are multi-word, context-required phrases and validation zeroed the accent-less alias weight—so the channel matches near-canonical forms rather than respellings: narrow and inspectable rather than broad and opaque.

7 Provenance and Error Analysis

The aggregate result is neutral; the question is what the 38 changes consist of.

7.1 Corrections are rule-consistent

Of the 23 corrections, 16 are HATE recoveries (7 from OFFENSIVE, 9 from CLEAN) and 7 are CLEAN de-flagging cases (6 from HATE, 1 from OFFENSIVE). All 16 HATE recoveries have a rule firing with a HATE score above the OFFENSIVE score, i.e. the rule points in exactly the corrected direction. The rule HATE score has median 0.7652 (mean 0.7507, range 0.2351–0.9939), so the evidence is strong, not marginal, for most of them.

All 7 CLEAN de-flagging cases have a silent rule ($r_H = r_O = 0$, so $\tilde{\mathbf{r}} = (0, 0, 1)$). With $\alpha_H = 0.2$ the rule term disappears from the HATE channel and $f_H = 0.2\, p_H$, so a

comment needs $p_H \geq 0.70$ merely to reach $\tau_H = 0.14$; one flagged HATE at $p_H \approx 0.5$ ends at $f_H \approx 0.10$ and the decision falls back to CLEAN, whose channel keeps the full neural weight ($\alpha_C = 1$). The rule never touches the CLEAN channel, so silence acts only as a multiplicative prior on HATE. We tested the alternative convention, where a silent rule contributes nothing and the neural scores pass through unchanged: it changes 30 decisions (10 for it, 14 against) and raises macro-F1 from 0.6602 to 0.6614 (95% CI $[-0.0015, +0.0042]$, $p = 0.44$; McNemar 14–10, $p = 0.54$). It is undecided, so we keep the shipped convention for its net corrections ($+8$ against $+4$).

7.2 Regressions are rule-driven and directional

The 15 regressions split into 10 rule-driven and 5 threshold-driven cases, and the split aligns perfectly with direction (Fig. 2b). In all 10 rule-driven cases (four OFFENSIVE→HATE, six CLEAN→HATE) the rule fires in the HATE direction, so fusion trusts lexical HATE evidence over the neural label and the lexicon is wrong about intensity or target. The OFFENSIVE cases are the expected failure of a phrase lexicon: profanity used as an intensifier rather than an attack. The CLEAN cases are false positives on non-abusive text. One is a political comment whose lexicon entry is a charged multi-word expression: the rule fires at $r_H = 0.62$, the transformer prefers CLEAN ($p_C = 0.515$ against $p_H = 0.473$), but the CLEAN channel misses $\tau_C = 0.60$ while the rule lifts f_H to 0.588, far above $\tau_H = 0.14$, so the decision goes to HATE. A lexicon entry's *political* sensitivity can therefore leak into moderation decisions (Sect. 8).

The 5 threshold-driven regressions are the mirror image: the rule is silent, the fused HATE score is $0.2\,p_H$, and when p_H falls below roughly 0.7 the HATE channel no longer wins; two miss by exactly 0.001 ($f_H = 0.1390$ against $\tau_H = 0.14$). The symmetry is the point: one mechanism ($\alpha_H = 0.2$, $\tau_H = 0.14$) produces 7 correct de-flagging decisions and 5 incorrect ones, and cannot be removed without also removing the corrections. The decomposition is exhaustive—26 of the 38 changes rule-driven, 12 threshold-driven—and that is what makes the hybrid explainable: not a heat-map over the transformer but a statement of which channel moved, in which direction, with which spans, and by how much.

7.3 The corrections fusion does not make

A channel with no signal and a channel whose signal fusion ignores look alike in an aggregate; the second is the case here. Among the 827 AI errors, 148 non-CLEAN comments have the rule firing in the correct direction and 151 CLEAN comments have the rule correctly silent, so the rule agrees with the gold label on 299 errors. Fusion acts on 23 of them (7.7%). The remaining **132 missed corrections** are non-CLEAN cases where the rule already points the right way and the hybrid is still wrong. Separately, 144 of the 151 silent-CLEAN errors stay wrong because the AI's HATE probability is high enough that even one fifth of it clears τ_H.

The 132 are not missed because the lexical evidence is weak: their median rule HATE score is 0.7609 (quartiles 0.67–0.85), indistinguishable from the 0.7652 median of the corrected cases. They are missed because the CLEAN channel keeps the full AI weight ($\alpha_C = 1.0$), so a confidently wrong CLEAN cannot be outvoted: the median fused CLEAN

score is 0.97 for missed cases versus 0.46 for corrected ones. Evidence-gated fusion (Sect. 8.4) would abstain here; we did not tune such a rule on test, which would invalidate the analysis.

7.4 What the rule channel responds to

We annotated the 16 HATE recoveries with a multi-label cue taxonomy: ten contain teencode or abbreviated profanity (`đ*o`, `dm`, `cmnr`), five standard-spelling profanity or an animal slur, and eight derogatory wording without profanity (*ngu*, *thua*, *nhục*); one has no surface cue. The categories overlap, so the counts sum to more than 16. The channel is therefore not merely a profanity detector: much of its value comes from derogatory wording a slur list would miss, and teencode is its largest single category.

Which components do the work? Re-tuning and re-freezing the fusion once per switch (Sect. 4.3 protocol), only one switch matters: removing the per-entry confidence costs 0.034 standalone macro-F1 (0.5258 to 0.4914), almost all of it on OFFENSIVE (F1 0.2389 to 0.0600), and no other switch moves the standalone engine by more than 0.008. Rescaling severity rewrites 1,046 of 6,680 rule scores yet changes no hybrid decision ($p = 1.0$): α and τ re-tune. Dropping the obfuscation views trades recall for precision (firings 1,052 $\rightarrow$ 779, precision on fired comments 58.75% $\rightarrow$ 69.58%); fuzzy matching never fires (0 matches), and case-folding cannot be ablated because all 6,680 test comments are already lower-case. Inside the hybrid no switch leaves noise ($|\Delta| \leq 0.0012$): the channel helps as a channel, not through its weights.

7.5 Case studies

Table 3 shows five decisions covering these mechanisms. Case 1 is the canonical rule rescue: two exact matches carry HATE from p_H=0.19 to f_H=0.81, past the 0.78 CLEAN channel the AI is confident in. Case 2 is the one correction in the five that the plain argmax would not make—HATE is 0.01 *below* OFFENSIVE, but OFFENSIVE falls short of its own τ_O=0.66 while HATE clears τ_H=0.14, which is the 0.0009 that per-channel thresholds buy (Table 2). Case 3 is a de-flagging: a zero-weight accent-less alias leaves the rule silent, the HATE channel collapses to $0.2\,p_H$, no channel clears its threshold and the fallback returns CLEAN. Cases 4 and 5 are the two failure modes of Sect. 7.2: an over-firing rule on profanity used as an interjection, and a HATE channel that misses τ_H by exactly 0.001 and de-flags a comment the AI got right.

8 Discussion

8.1 What the hybrid is good for

The result is better read as a successful audit: 38 of 6,680 decisions change, each described by a channel, a direction and a mechanism.

Triage and accountability. A queue can sort comments by whether the decision was rule-supported: a HATE decision with a matched span and $r_H = 0.8$ is easier to review than one with no lexical evidence, and the span, the entry and the arithmetic are what a platform can defend in policy terms.

Table 3. Case studies from the 38 changed decisions (∗ masks profanity). $H/O/C$ are the fused channel scores, τ_c the frozen per-channel thresholds and r_H the aggregated HATE rule score; the figure after each span is its per-match score. Rows 1–3 are corrections, 4–5 regressions.

#	Comment and rule evidence	Gold → AI → Hybrid
1	*để avata bít ngay là ngu hơn chó*	HATE → CLEAN → HATE
	r_H=0.97; ngu hơn 0.86, chó 0.79 (exact)	H 0.81 > C 0.78
2	*... cũng đ*o thể yêu thương nổi*	HATE → OFFENSIVE → HATE
	r_H=0.71; cũng đ*o 0.71 (exact)	H 0.49 < O 0.50
3	*hóng bro tét đ*t thằng tùng*	CLEAN → HATE → CLEAN
	r_H=0; accent-less alias, weight 0	H 0.10 < τ_H=0.14
4	*... nói không.. đợi cái loz*	OFFENSIVE → OFFENSIVE → HATE
	r_H=0.87; cái loz 0.87 (exact)	H 0.70 > O 0.68
5	*...abe tuổi lol mà so đc*	HATE → HATE → CLEAN
	r_H=0; no scoring match	H 0.1390 < τ_H=0.14

Editing and diagnosis. Adding a slur variant is a configuration change, not a retraining run, and an edit affects the HATE channel (weight 0.8) far more than the others. The decomposition exists only because the channels are separate: a monolithic classifier improving by 0.0027 macro-F1 would give no handle on whether the change was lexical or neural.

8.2 Why accuracy does not move

First, *coverage is low*: the rule fires on 15.75% of comments, so a perfect rule channel could still only change decisions in that subset. Second, *the majority class dominates*: with 83% CLEAN, only the 1,132 toxic comments can improve, and 38 changes is 3.4% of that support—enough to matter operationally, too little to move a macro-average beyond its confidence interval. Third, *tuning is safety-oriented*: protecting CLEAN and gating HATE on lexical support produces both the de-flagging and the missed corrections. Neutrality is a *choice*, not an absence of signal.

8.3 The directional asymmetry is a policy question

In a binary "remove / keep" framing the asymmetry of Sect. 7.2 is a recall-oriented tradeoff; in a three-class framing it is not, because an OFFENSIVE comment mislabelled as HATE can trigger a stronger sanction than the content warrants, and a political comment mislabelled as HATE is a different kind of harm.

That false positive matters: the entry is a charged multi-word expression labelled HATE from co-occurrence, so the lexicon is not a neutral artefact—it encodes editorial judgments that enter at weight 0.8. Deployments should separate *profanity* from *political* entries and weight them accordingly.

8.4 Design implications

Four changes follow. (i) **Evidence-gated fusion**: when the rule fires strongly against a high-confidence CLEAN prediction, abstain or escalate. (ii) **Separate thresholds per mechanism**: requiring $r_H \geq 0.5$ before the HATE channel can override would make the rule-driven and threshold-driven paths separately tunable. (iii) **Per-class calibration** of

the CLEAN channel, which currently blocks the missed corrections. (iv) **Human-in-the-loop** for the 804 both-wrong comments, which no fusion of these two channels can fix.

8.5 Limitations

Single benchmark and language. All results come from ViHSD and Vietnamese. The lexicon and cue taxonomy are language-specific; the fusion protocol is not, but we have not tested it elsewhere.

One seed per backbone. Each backbone was trained once with seed 42, so neural variance is not estimated and the 0.0027 difference is well inside what seed variance could produce: the tests rule out a large effect, not a small one.

The rule engine is weaker than designed. Validation set the accent-less alias weight to zero and the negation factor to one, removing the main defence against dropped diacritics and the discount for negated slurs visible in the OFFENSIVE-as-intensifier regressions.

Explanations are not evaluated by humans. The evidence record is faithful for the lexical channel and decomposes the changed decisions, but we do not measure whether reviewers find it useful, and the cue taxonomy is a post-hoc annotation.

8.6 Ethics

The comments are public ViHSD test comments and are masked as described in Sect. 3. The system is decision support, not automatic sanctioning: the directional asymmetry and the political false positive show that automatic HATE labels from this pipeline should be reviewed before they affect a user. The lexicon should be treated as a governed artefact with a change log, because its entries shape moderation outcomes.

9 Conclusion

We attached a 668-term Vietnamese hate-speech lexicon to four multilingual transformer backbones as an evidence channel and evaluated it on the ViHSD test split. On the focused backbone the hybrid improves macro-F1 from 0.6575 to 0.6602—a net of eight comments, with paired-bootstrap $p = 0.29$—so we do not present it as an accuracy contribution.

The contribution is the provenance of the change. All 23 corrections are rule-supported (16 HATE recoveries with median rule score 0.77, and 7 CLEAN de-flagging cases where the rule is correctly silent), and all 10 rule-driven regressions push towards HATE, localising the failure mode to an aggressive channel rather than to coverage. A further 132 missed corrections show the rule already pointing right while a full-weight CLEAN channel blocks the fix.

The broader lesson is methodological: a hybrid evaluated only by its aggregate metric looks like a null result, whereas the same hybrid decomposed by channel and direction yields an auditable decision procedure and a design agenda. Future work: validate evidence-gated fusion, separate profanity from political entries, restore negation and accent-less matching, and measure whether the evidence record helps reviewers.

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